

Kinematics and Dynamics of Machines MEL3020

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Velocity Analysis using instant centers

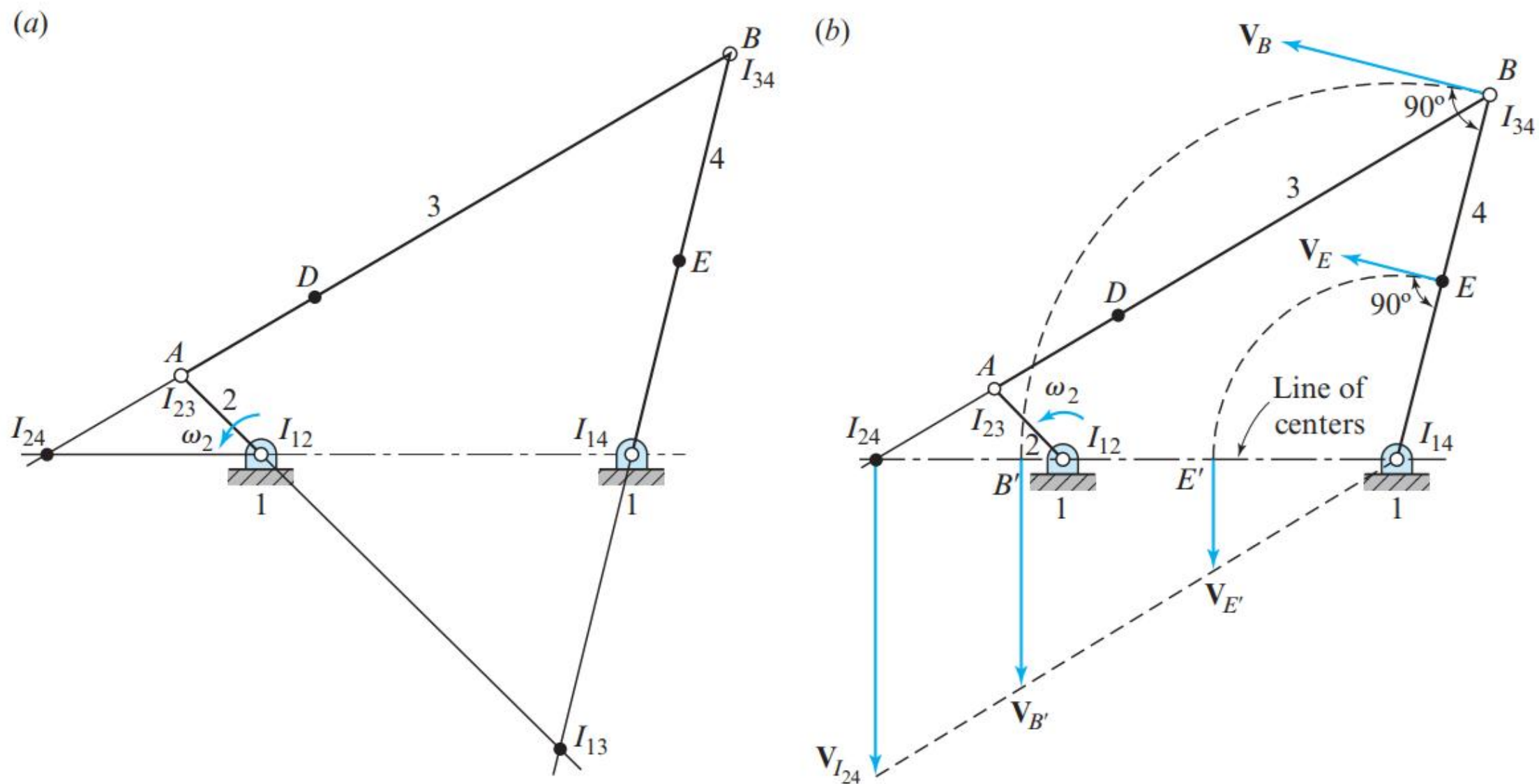


Figure 3.29 Graphic velocity analysis using instant centers.

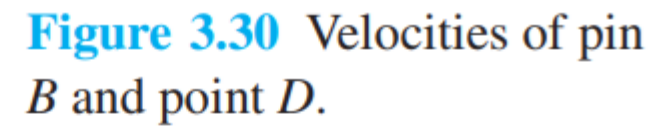


Figure 3.30 Velocities of pin B and point D .

EXAMPLE 3.12

For the device shown in Fig. 3.31, some of the links are visible, whereas other links are enclosed in a housing. However, the location of instant center I_{25} is known, as shown in the figure. If the velocity of point C of link 6 is $\mathbf{V}_C = 10$ m/s to the right, determine the angular velocity of link 2.

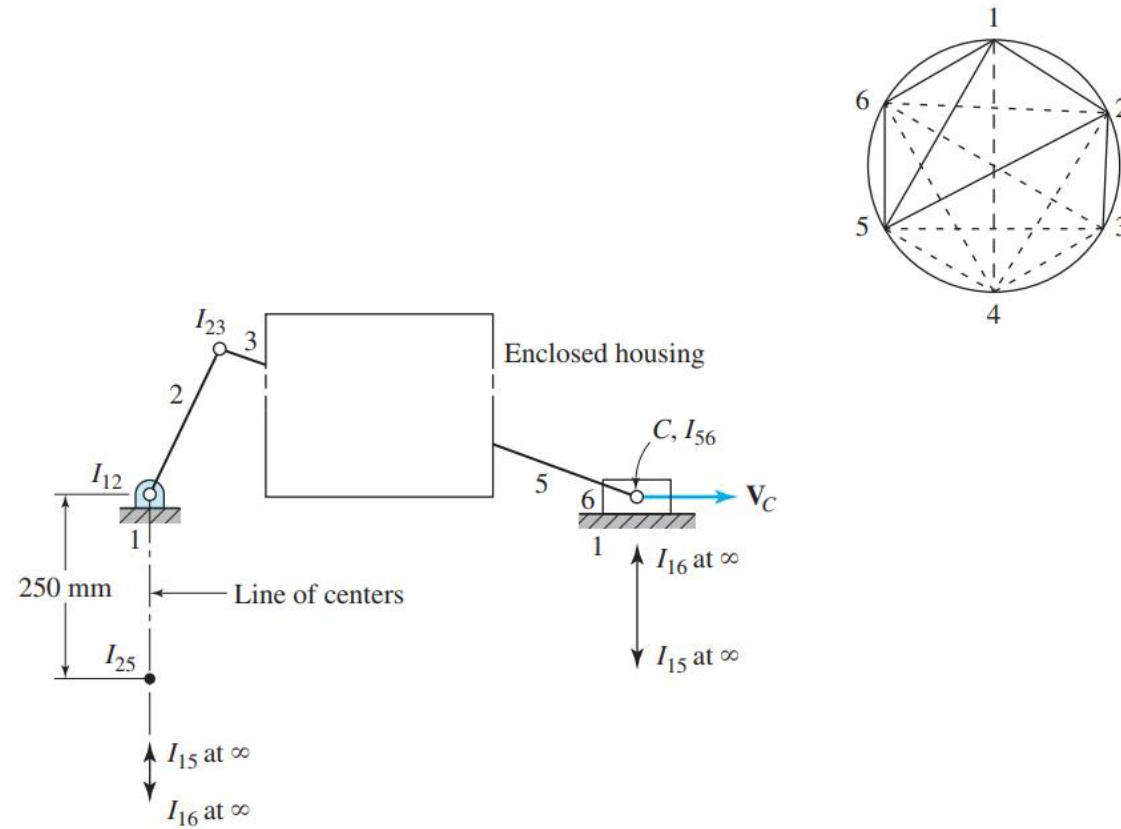
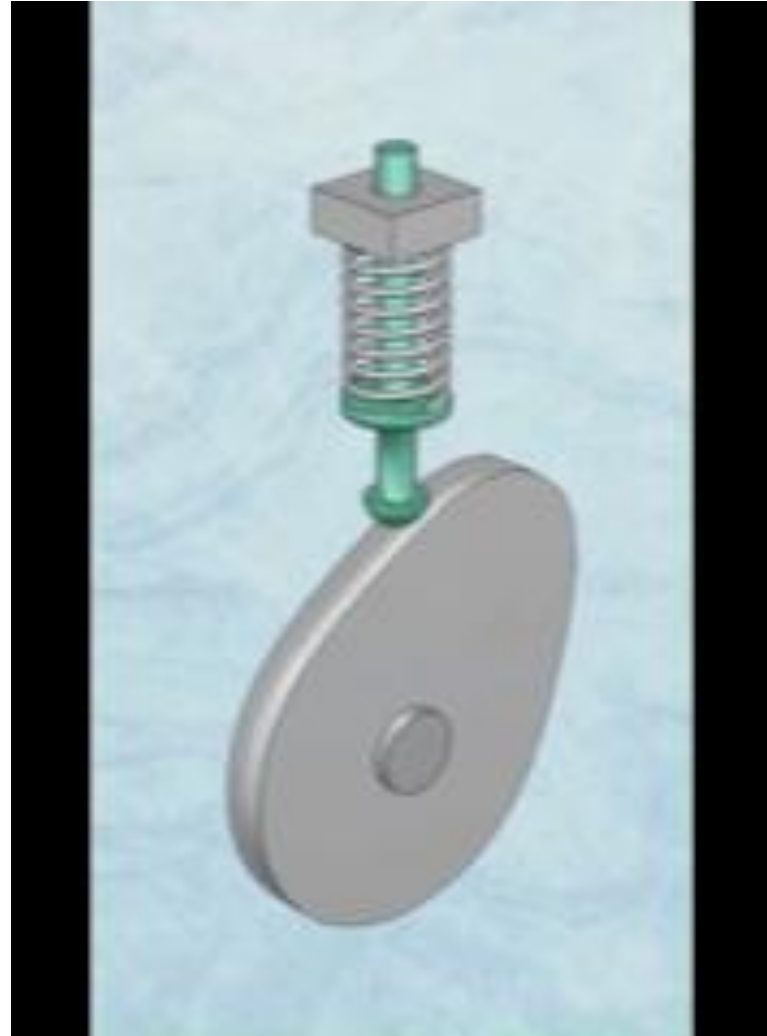
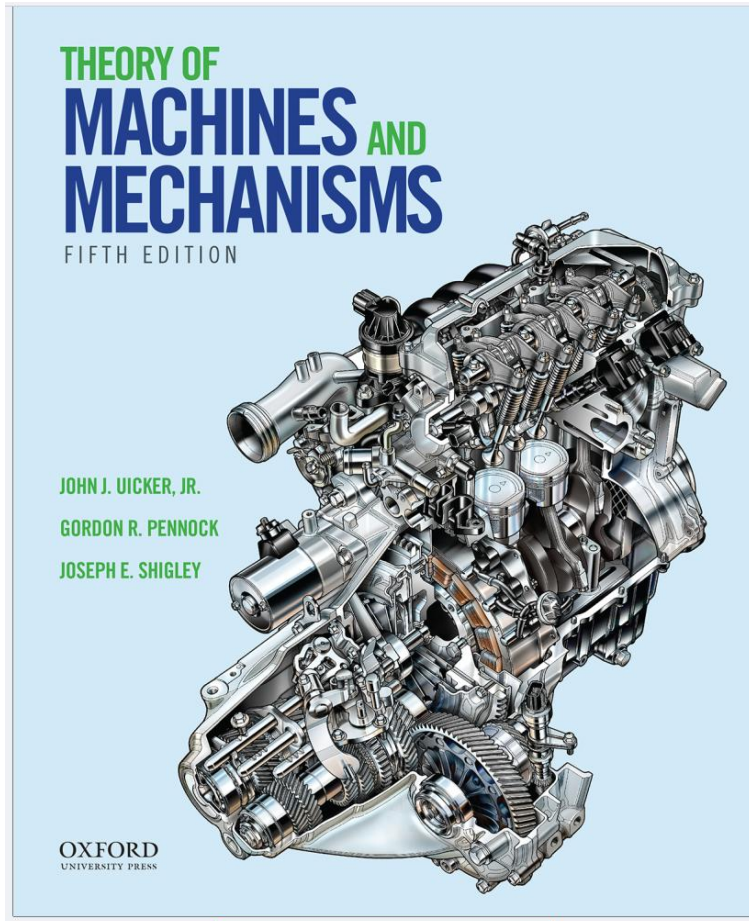


Figure 3.31 Links enclosed in a housing.

CAM Analysis



CAM Classification

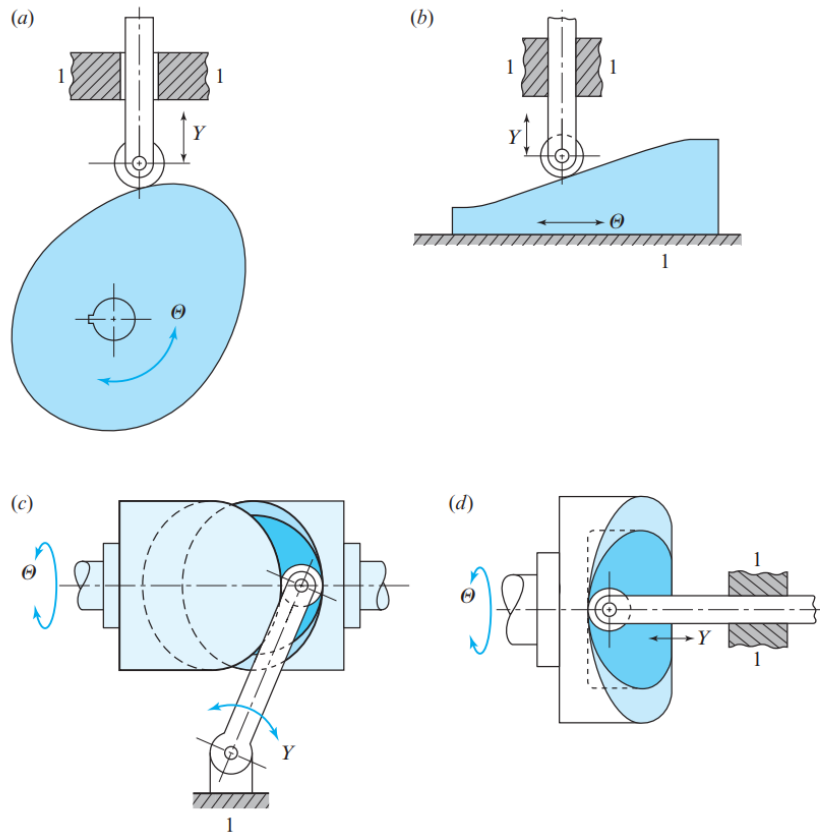


Figure 6.1 (a) Plate cam; (b) wedge cam; (c) barrel cam; (d) face cam.

CAM Classification

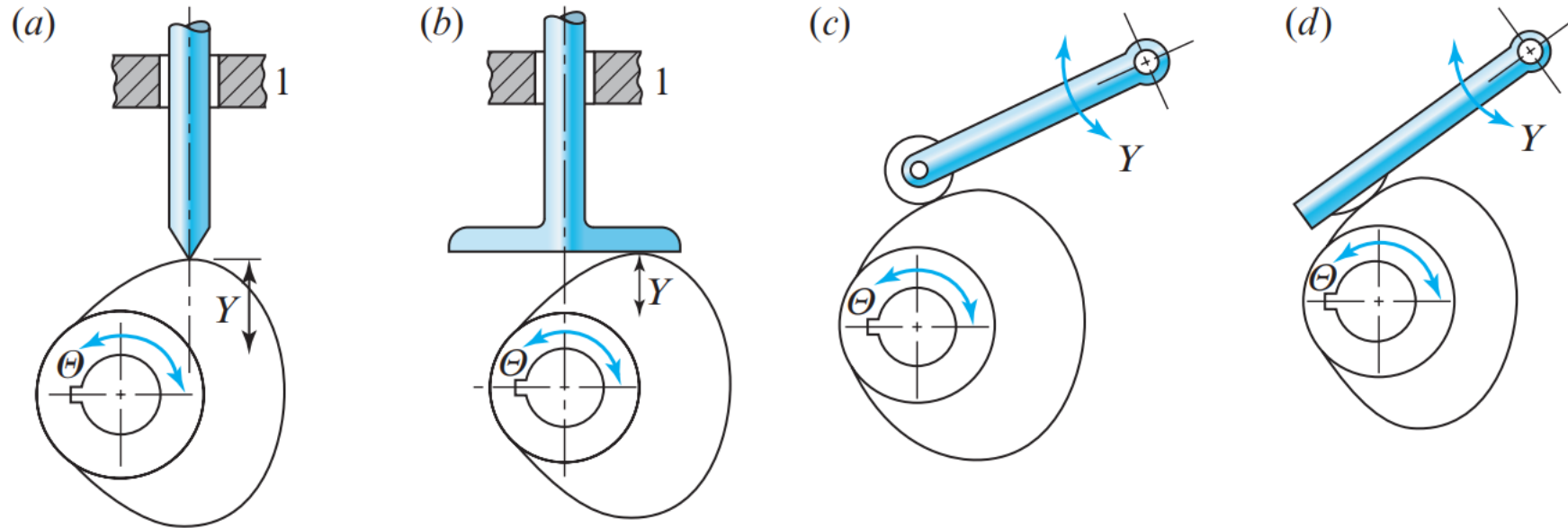


Figure 6.2 Plate cams with (a) an offset reciprocating knife-edge follower; (b) a reciprocating flat-face follower; (c) an oscillating roller follower; (d) an oscillating curved-shoe follower.

CAM Classification

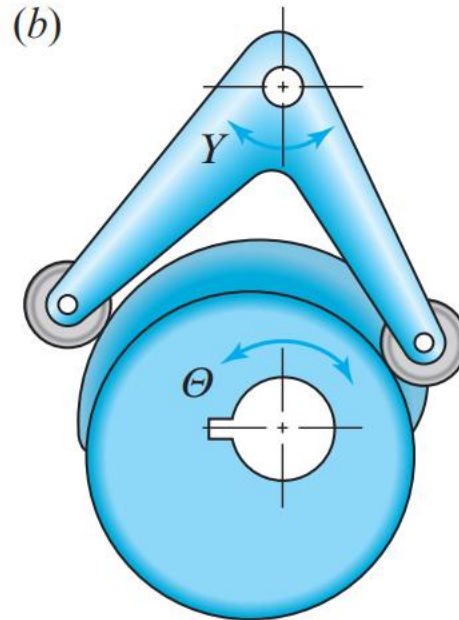
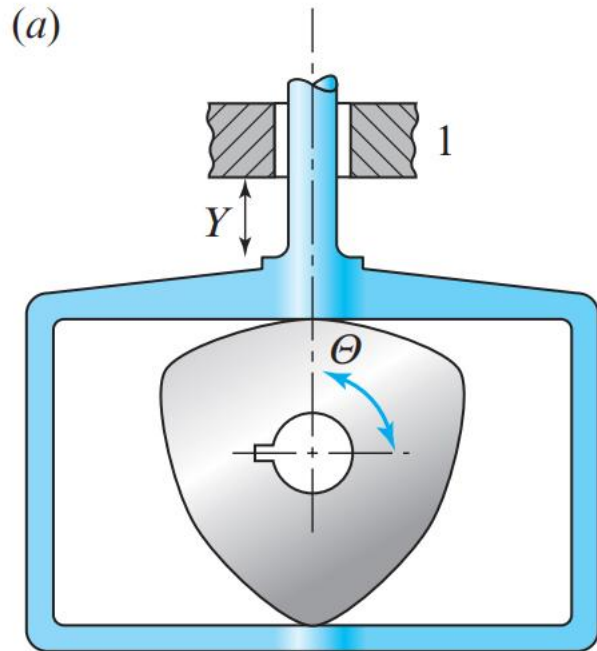


Figure 6.3

(a) Constant-breadth cam with a reciprocating flat-face follower; (b) conjugate cams with an oscillating roller follower.

Displacement Diagrams

During the rotation of a cam through one cycle, the follower executes a series of events as demonstrated in graphic form in the *displacement diagram* of Fig. 6.4. In such a diagram, the abscissa represents one cycle of the input, Θ (usually, one revolution of the cam), and is drawn to any convenient scale. The ordinate represents the follower travel, Y , and, for a reciprocating follower, is usually drawn at full scale to help in the layout of the cam. On a displacement diagram, it is possible to identify a portion of the graph called the *rise*, where the displacement of the follower is away from the cam center.

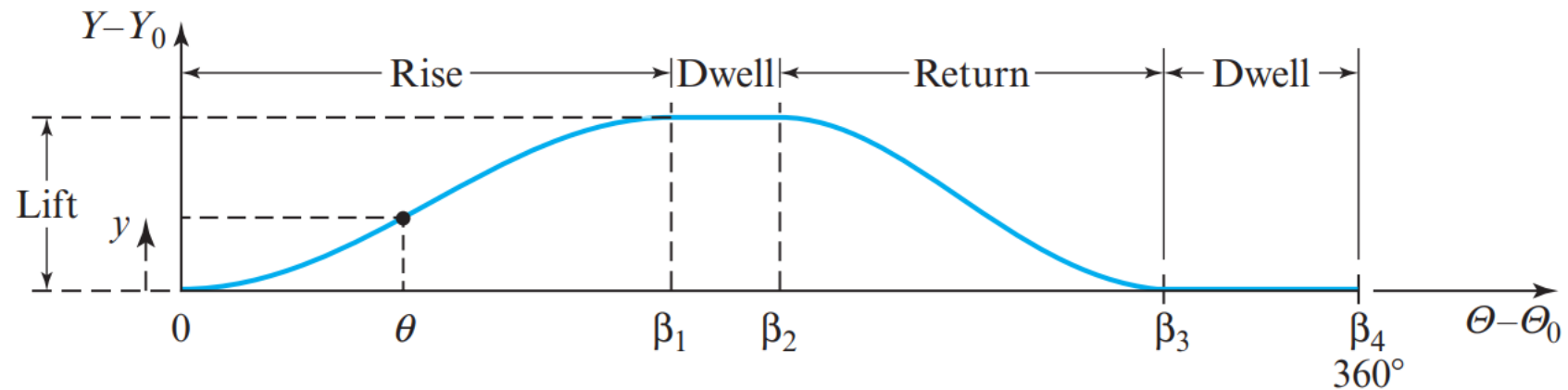


Figure 6.4 Displacement diagram for a cam.

Displacement Diagrams